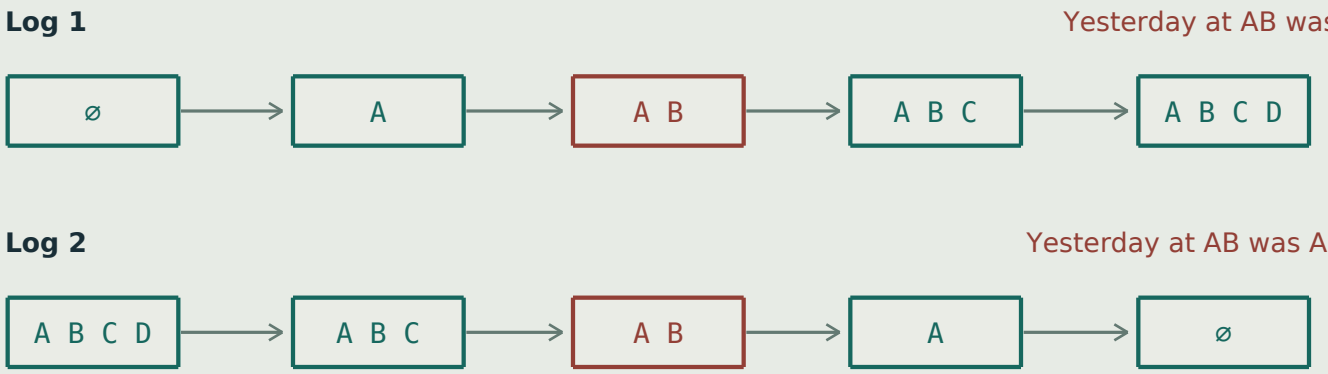
Which plate was yesterday?

Adversarial reading of The Counterseal · first formulation

At the plate AB the reader can see two neighbors, A and ABC. The Counterseal identifies A as yesterday. An auditor can give exactly the same reader, in exactly the same fictional world, a visit for which the correct answer is ABC.

TWO AUDITOR LOGS / ONE READER VIEW



Arrows show the auditor’s visit order. The reader receives the path up to reversal, with AB distinguished. Every mark name and both of its faces agree.

The correct answer changes when the auditor reverses the visit. The input to the decoder stays identical. Every deterministic decoder therefore fails on at least one log. If the auditor chooses the two logs with equal probability, even a randomized decoder succeeds at most half the time.

Disposition. Retire the universal claim that the catalogue can identify yesterday. The concrete prediction that both A and ABC must be lawful survives. It follows directly by taking the intersection and union of AB and AC.

Chronology enters this comparison through an external auditor’s scoring rule. Neither visit was observed in nature. The counterexample challenges an entailment inside an explicitly fictional world.

Give the reader more

Distinguishable ends, fault detection and a readable history

Distinguishable ends

Suppose A has three independent children B, C and D. The plates are the blank plate together with A joined to any subset of its children. The blank plate has one neighbor. The full plate has three. No symmetry interchanges the ends. The very same two logs on p. 1 are lawful, still have identical reader views, and still have opposite answers.

Property of the catalogue	Diamond	Fork
Lawful plates	6	9
Increasing paths from blank to full	2	6
Neighbors at blank / full	1 / 1	1 / 3

A real benefit of the second face

A single faulty bit on either face breaks the match. A one-bit discrepancy can therefore be detected. Between any two neighboring lawful plates, however, a half-written seal sits one bit away from each. Reading this inconsistency cannot tell which plate was written last. The minimum Hamming distance of the paired inscriptions is two.

What one external bit would buy

For a fully given nested path with precisely its two global orientations in play, one trusted mark of which end began the visit is sufficient and necessary. It selects one of the two logs. Even granting that a visit grew from blank to full, the diamond has two increasing histories: B can precede C or C can precede B. Knowing only the endpoint plates leaves that order open.

A new law allowing only additions would also select an orientation. Its explanatory cost is explicit: irreversible growth would then be a further premise.

What a catalogue determines

A proof for any finite number of named marks

The surviving statement

Let F be a finite family of plates containing the empty and full plates, closed under union and intersection, and separating names. Define x ≤ y if every plate containing y contains x. Then ≤ is a partial order and F is exactly its family of downward-closed subsets.

Proof

Reflexivity and transitivity are properties of logical implication. If x ≤ y and y ≤ x for distinct names, the two names occur together in every plate. This violates separation. Every plate is therefore a downward-closed set in the recovered order.

For each y, take the intersection I(y) of all plates containing y. This family is nonempty because it contains the full plate. Finite intersection closure makes I(y) a lawful plate, and its members are exactly the names x with x ≤ y. Every downward-closed set is the union of these I(y) over its members. Finite union closure makes that union lawful. The empty set is already in F. This proves both directions.

The held-out plates

Among the 16,384 four-name families with the empty and full plates included, exactly 219 satisfy the closure and separation rules. Twenty-five contain both AB and AC. Every one of those twenty-five contains A and ABC. Those two positive readings suffice for this prediction, while the full set of prerequisites can differ between catalogues.

This is the finite order/topology correspondence in downset convention. Knowing every member of a catalogue in the story is a strong access assumption. A handful of positive readings does not certify that a prerequisite has no counterexample.

Where the disproof stops

Reversible writing, prior work and the next draft decision

A reversible realization

For each mark i, toggle both faces if the resulting plate is lawful. Otherwise leave the plate unchanged. This map swaps disjoint pairs and fixes the remaining plates, so applying it twice is the identity. Every step in the two auditor logs can be realized this way. The model grants neither a distribution of logs nor a preferred initial plate.

Two external controls

Bennett showed how a computation can retain intermediate information in a logically reversible construction. Mlodinow and Brun study conditions under which robust physical memories align with a thermodynamic arrow. Both bear on the distinction between storing a distinction and explaining an orientation. The fictional plate rules contain no bath, entropy gradient, robust physical preparation or boundary condition.

The order-reconstruction proof uses the standard correspondence between finite T_0 spaces and partial orders. Its role here is to show exactly how much structure the reader really can recover.

Disposition for the next formulation

Archive the temporal claim as disproved. Keep the prerequisite theorem as a positive control. A future fictional candidate may posit an independently readable directional asymmetry. It must state who can read it, how it changes under reversal, and what would count as its absence before it can make a new wager.

C. H. Bennett, IBM Journal of Research and Development (1973), “Logical Reversibility of Computation.”

L. Mlodinow and T. A. Brun, Physical Review E (2014), “Relation between the psychological and thermodynamic arrows of time.”

A. Fix and S. Patrias, “Enumeration of Homotopy Classes of Finite T_0 Topological Spaces,” §2.